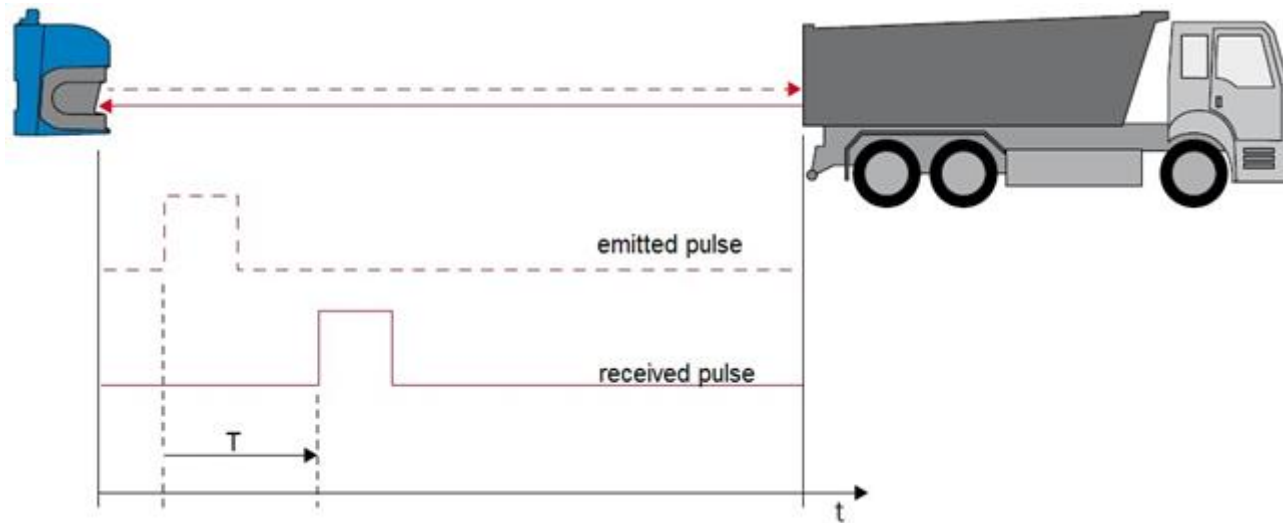


LIGHT DETECTION AND RANGING (LiDAR)

WORKING MECHANISM

❑ A pulsed laser beam can be used to measure distance.

The pulse of coherent light is emitted in a particular direction, thus the reflected signal is measured, to evaluate the delay of the reflected pulse.



The delay T is known as “Time of Flight” (ToF)

which would be $T = \frac{D}{c}$; being D the travelled distance, and c the speed of the light.

The travelled distance D , is twice the distance to the reflecting surface.

(image from <https://www.sick.com/au/en/detection-and-ranging-solutions/2d-lidar-sen>)

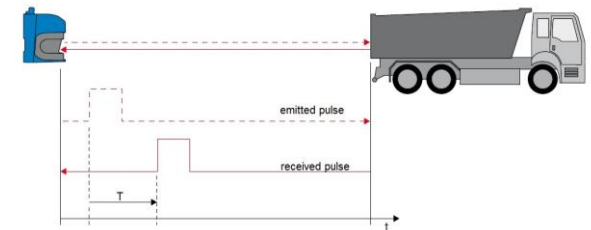
We find some technical issue here (as it usually happens)

The ToF must be measured: hardware is not perfect; it introduces errors in the measurement of the ToF.

To put this in context: $c \sim 300,000 \text{ km/s} = 3 \times 10^8 \text{ m/sec}$

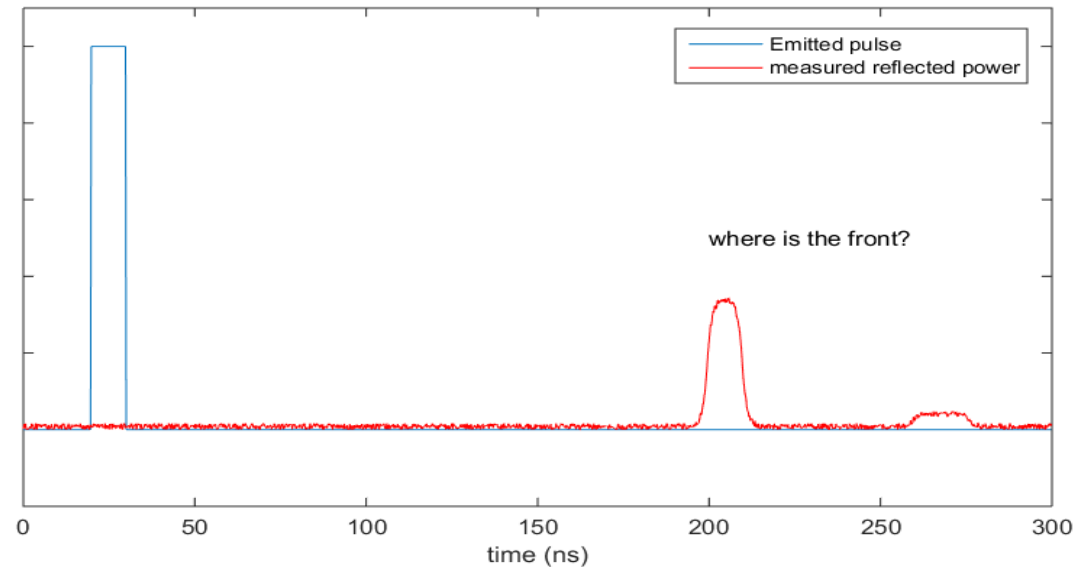
$L = 1 \text{ m} \rightarrow T = (1 \text{ m})/c \sim 3.3 \text{ ns}$.

$L = 100 \text{ m} \rightarrow T = (100 \text{ m})/c \sim 333 \text{ ns}$.



If we want an accuracy of 1 cm: $dL = 1 \text{ cm} \rightarrow dT \sim 0.034 \text{ ns}$ (<4% of a ns!)
(you would not do that using an Arduino loop!)

The reflected signal is polluted by noise, and by other echoes, and it is also distorted, and that distortion depends on the relative orientation/shape of the spot of surface being illuminated by the laser beam.



And from that received analog signal, the hardware measures/estimates the ToF, with an accuracy of 0.04ns, for achieving an accuracy of 1cm.

And yet, a common commercial LiDAR can achieve cm accuracy, and ranges of 100m.

(I have exaggerated the spread of the distorted reflected pulse, in the plot. I think its flanks should evolve in intervals of about 3ns, based on some estimations I tried.)

The spread of the flank increases with the measured distance and with the relative orientation of the surface at the point at which the beam hits the surface.

(but that is not our business , in MTRN4010)

- ❑ A pulsed laser beam can be used to measure distance.

The pulse of coherent light is emitted in a particular direction and the reflected signal is measured, to evaluate the delay of the reflected pulse.

- ❑ The propagation time is used to estimate distance (as the speed of EM waves is a well-known constant).
- ❑ This process can be repeated tens to hundred of thousands of times /second (in theory even millions!).
- ❑ If the same process is repeated at different directions, a surface can be scanned.
- ❑ Typical laser scanners operate in a plane of scanning (2D).
- ❑ More advanced ones do scan in 3D (2D angles, elevation and azimuth)

- ❑ This process can be repeated tens to hundred of thousands of times /second (in theory even millions!).

Millions? Why? (where is that limit from?)

For measuring distances up to $<100\text{m}$, the reflection would take, in the worst case ($L=100\text{m}$), a ToF $T=100\text{m} / c = 333.4\text{ns}$. As we measure the reflection propagation time too, $\text{ToF}=0.67$ microseconds

→ in theory, we can emit and read the pulse, more than 1 millions times / second !

→ Let's put it in context again a camera of 3 megapixels can give the 3 millions reads @ a rate of 30 times /second ($30 \times 3\text{millions}/\text{sec}$).

→ But a rgb camera cannot give range.

2D LiDAR

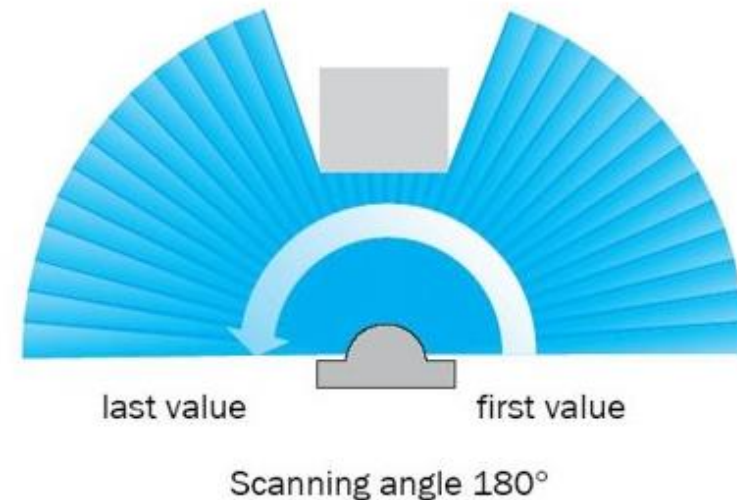
Here, a 2D LiDAR scanning 180 degrees (FoV = 180°), in a plane

The laser beam is pulsed periodically, at different directions (in its azimuth).

This is usually done by using a rotating mirror to deflect the emitted beam to the desired directions.

There is a good number of different brands and models

Typical FoV: ,100°,180°, 270° , and also 360°; depending on brand/model, and mode of operation chosen)

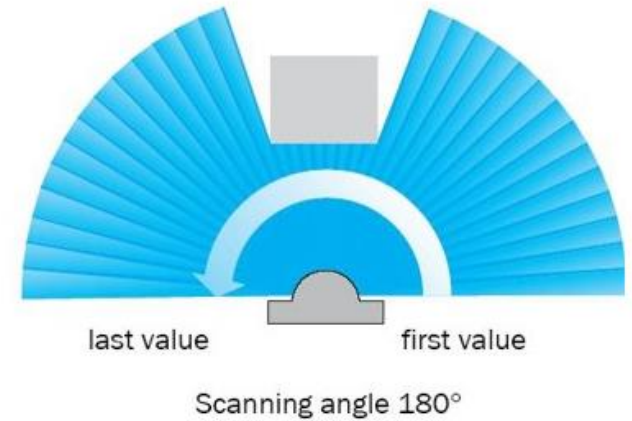


Angular Resolution:

In addition to the FoV, the angular resolution is a relevant parameter.

Usual resolutions : 1° , 0.5° , 0.33° , 0.25° , 0.1°

The one we will see soon in the lecture, operates at 0.33° (3 beams / degree)



WORKING MECHANISM

LiDARs are “Active” sensors.

Advantage: Do not require external illumination (as a camera would need), so they work well in total darkness as well

Disadvantage: may interfere with other active sensors operating in the same range of wavelengths.

It may need to compete with natural sources of light in the same IR spectrum range (sun)

LIMITATIONS

LiDAR are based on light to see objects, which may result in:

It may need to compete with natural sources of light in the same IR spectrum range (sun)

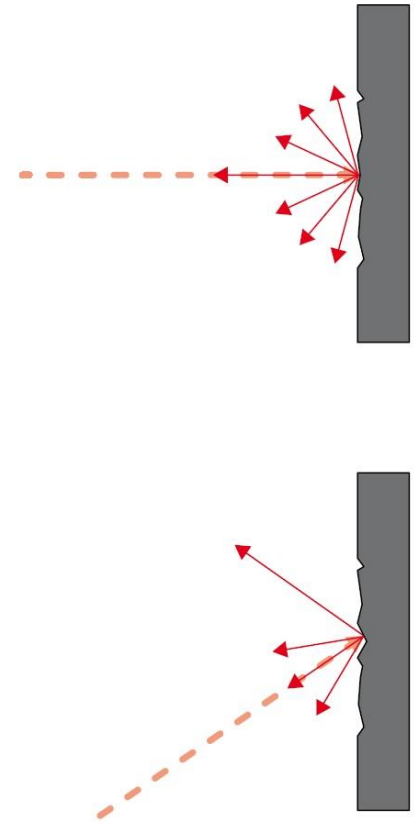
Occlusions , as IR light suffers effects like visible light (radars would not)

Environmental conditions (dense dust pollution, fog, rain)

LIMITATIONS

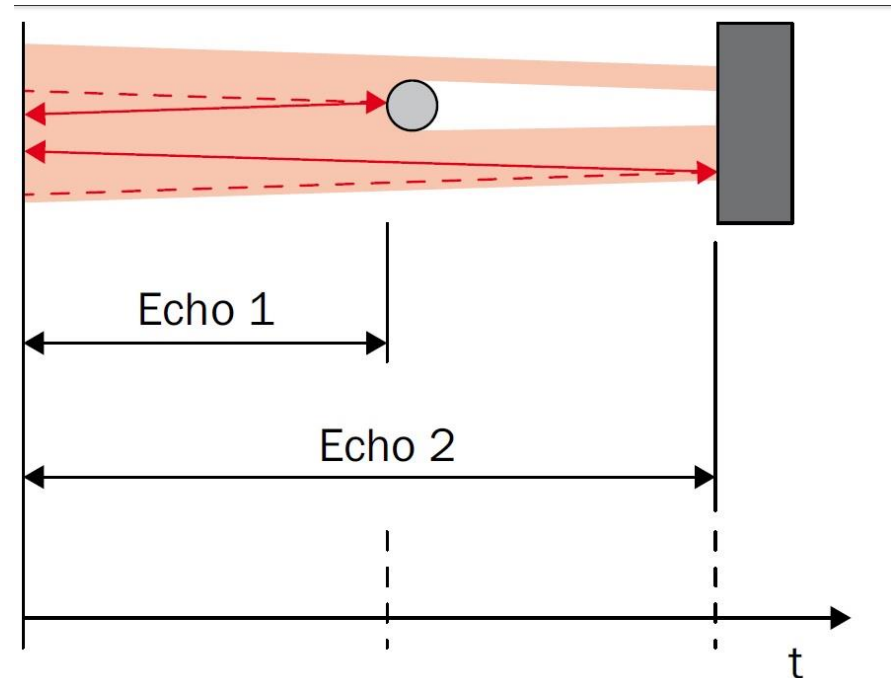
Effect on reflected light, due to surface characteristics

Not all the surfaces are good reflecting light in the incident direction (if not enough energy is returned, no pulse detection will occur).



LIMITATIONS

Multiple echoes

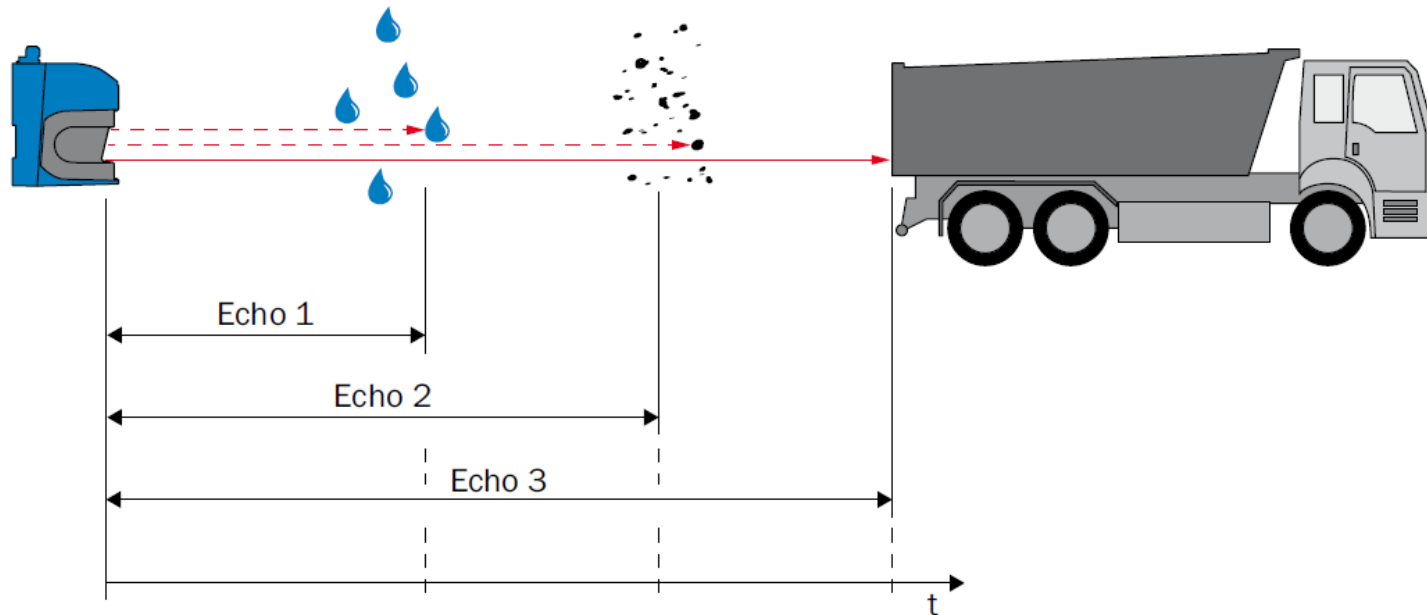


The laser beam is not a line of infinitesimal section. It has certain thickness (which increases with the travelled distance, due to laser divergence)

The beam may partially hit multiple surfaces, resulting in multiple echoes
(proper processing may see a benefit in that!)

LIMITATIONS

Multi-Echo may happen due to environmental conditions (rain, fog, dust in suspension)



Those “partial obstacles” still allow transmission, however producing more attenuation of the light (reducing maximum range of detection) and also generating additional reflections with result in echoes (which our processing software need to infer).

(image from <https://www.sick.com/au/en/detection-and-ranging-solutions/2d-lidar-sen>)

LIMITATIONS

→ Resolution is angular.

The size of the beam, and the (linear) density of points (points/meter) varies as a function of the distance, $dS = da \cdot \pi / 180 \cdot Range$

In which **da** is the angular resolution (expressed in degrees, usually da=1 degree or 0.5 degrees or 0.25 degrees)

Higher angular resolutions are offered, but those may be achieved by “interleaving” , which means that multiple consecutive scans are performed having the max real resolution (e.g. 0.25 degrees) and having small shifts between scans (e.g. 0.25/2), so that 2 scans combined achieve 0.25/2 of resolution.

This is usually OK when both, sensor and objects, are static.

REFLECTIVITY

Different materials have different reflectivity, which may generate issues.

- Highly reflective surfaces may saturate some laser detectors. Some dark/opaque surfaces may absorb too much of the light, resulting in reflections that are not intense enough to be measured by the detector.
- Minimum detectable object size depends on reflectivity (narrow but highly reflective could still reflect enough energy to be detected)
- A strong sunlight reflection off a highly reflective areas may "saturate" a receiver, affecting its performance.
- Reflectivity of scanned surfaces affects maximum range of the sensor.

REFLECTIVITY

Here, we can see a table which describes reflectivity indexes of typical materials.
(from maker of industrial LiDARs, RIEGL)

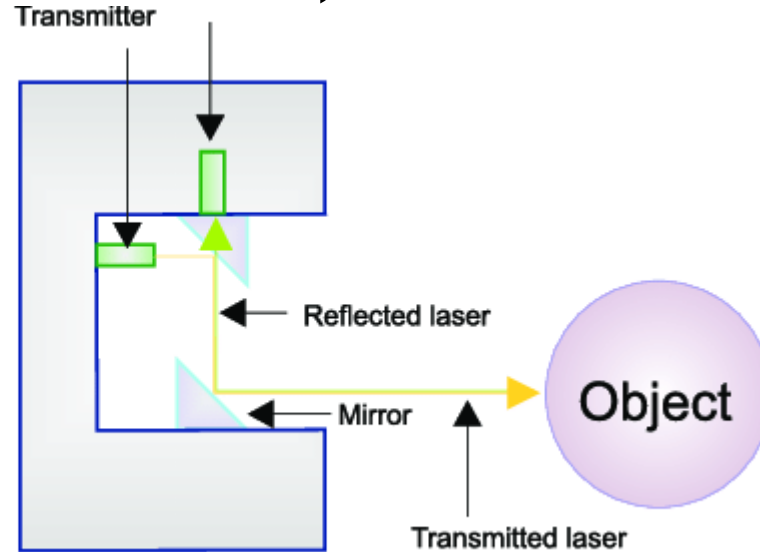
MATERIAL	REFLECTIVITY @ $\lambda = 900 \text{ nm}$
Dimension lumber (pine, clean, dry)	94%
Snow	80-90%
White masonry	85%
Limestone, clay	up to 75%
Deciduous trees	typ. 60%
Coniferous trees	typ. 30%
Carbonate sand (dry)	57%
Carbonate sand (wet)	41%
Beach sands, bare areas in desert	typ. 50%
Rough wood pallet (clean)	25%
Concrete, smooth	24%
Asphalt with pebbles	17%
Lava	8%
Black rubber tire wall	2%

Source: www.riegl.co.at

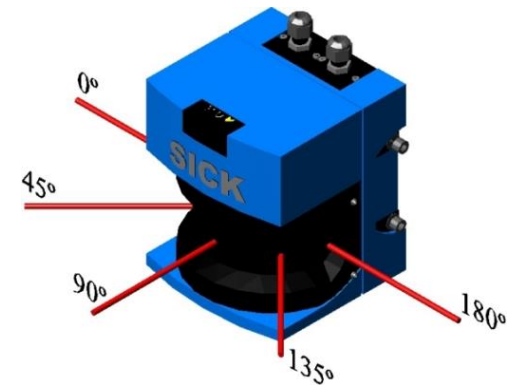
DUST, SMOKE, VAPOR, RAIN?

- Laser measurements can be degraded by interacting with vapor and also with dust and smoke particles, which scatter the laser beam, and do also attenuate the signal returned from the target.
- Using “last-pulse” measurements can mitigate certain problems with this interference (emitted pulse results in multiple reflections, so that the last one is assumed to be the “real one”)
- Systems that are expected to work in such conditions regularly can be optimized for these environments (in addition certain processing techniques which allow to properly extract information of the polluted/distorted measurements; such as Bayesian estimation, CNN, etc.)

EXAMPLE: 2D SICK LMS 200 (A CLASSIC!)



<https://www.sick.com/au/en/detection-and-ranging-solutions/2d-lidar-sensors/lms2xx/lms200-30106/p/p109843>



Many variants

Industrial grade

Example: LMS 200 family

- Range ~ 80 meters
- Resolution : 1 cm
- Angular range : 100 - 180 degrees
- Angular resolution : 0.25, 0.5 or 1 degrees
- Scan time : Typical = 13.5 milliseconds (~74 fps)(*).



LMS200



LMS211



LMS221

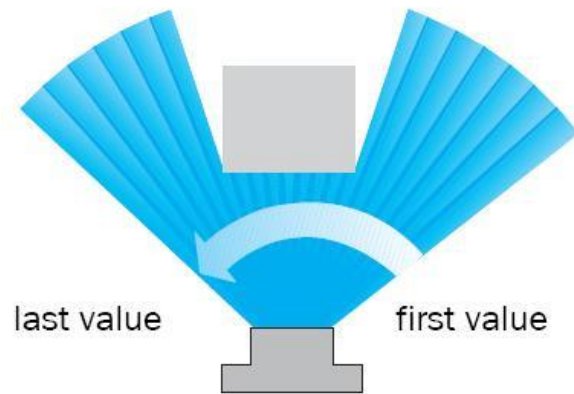


LMS291

There are other models/brands that offer better angular resolution and more range.

(*) E.g. it scans 180 degrees, at a resolution of 1 degree in 13.5ms.

LMS211

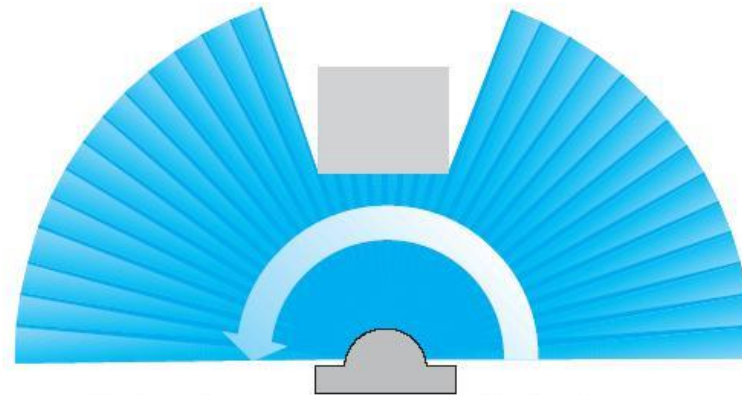


last value

first value

Scanning angle 100°

LMS200/LMS221/LMS291



last value

first value

Scanning angle 180°

(we see a demonstration in class , small LMS
TIM, (FoV) = 225 degrees)

Industrial grade.

Can work in harsh contexts
(e.g., mining)

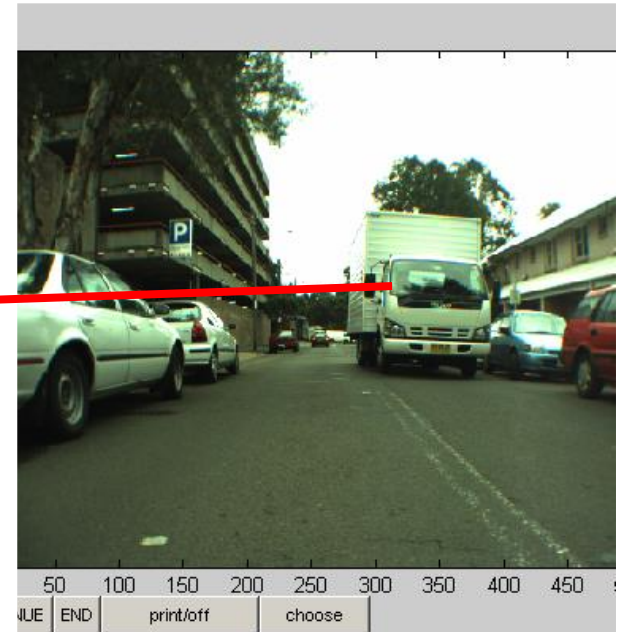
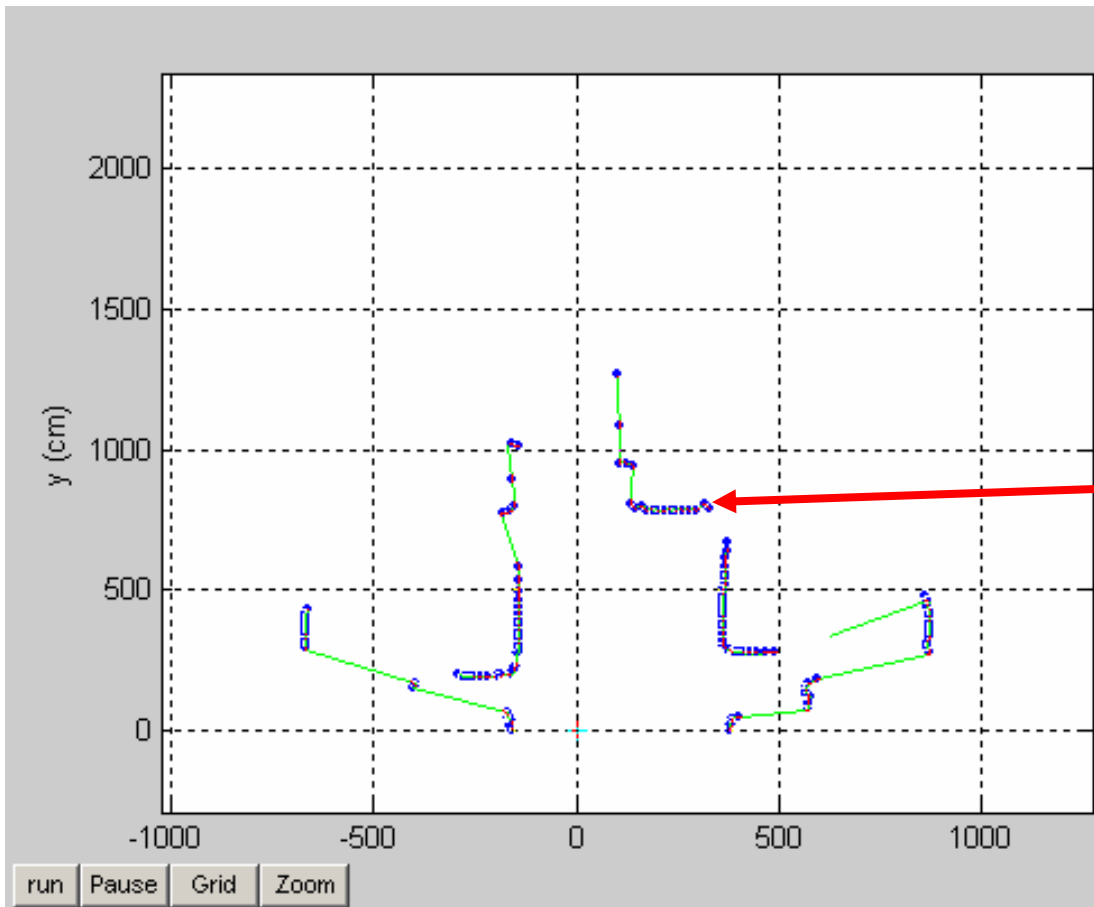
(lecturer had experience using them in
those contexts)



Laser scanner installed as part
of collision avoidance system.



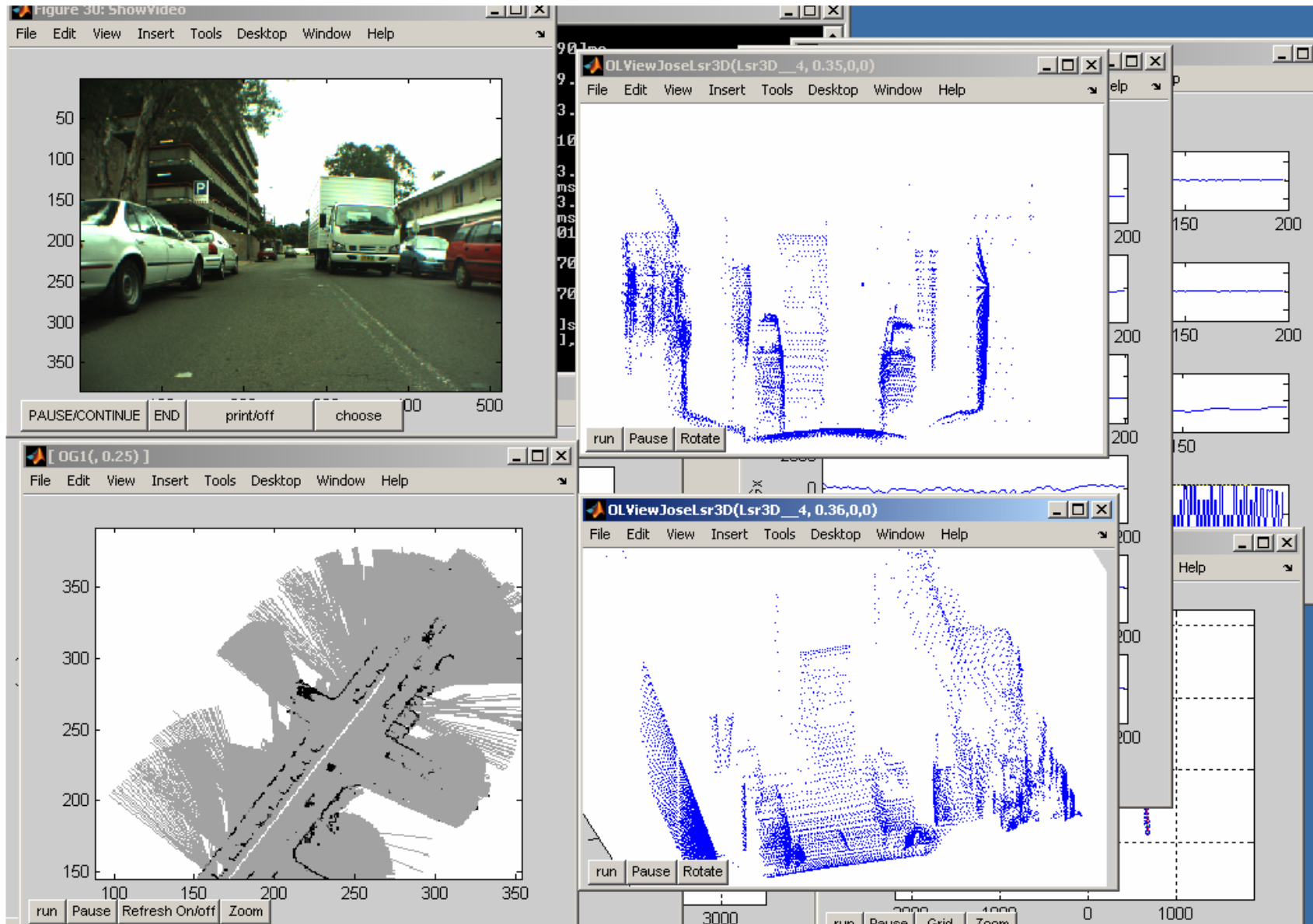
... as part of truck guidance system.

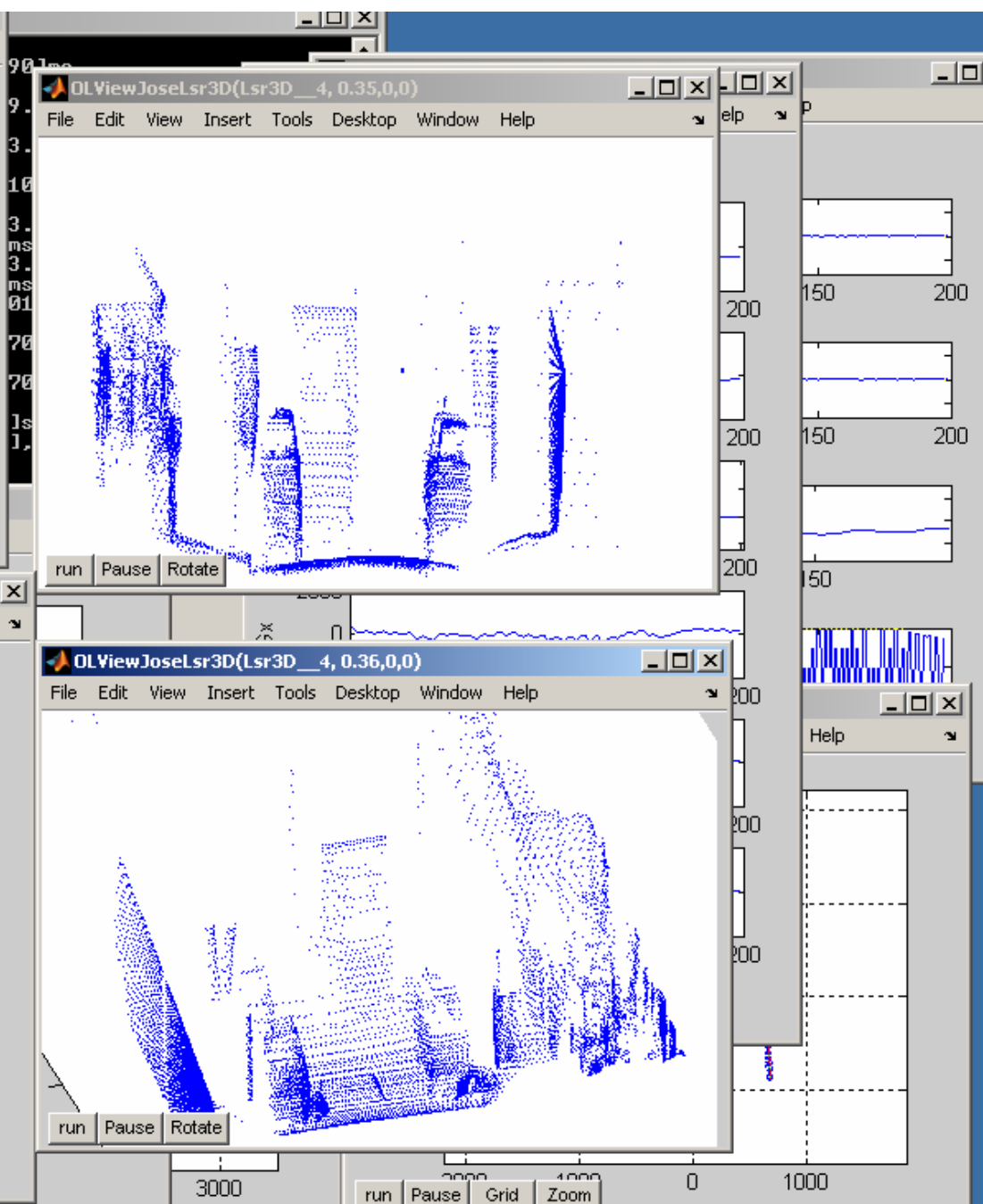
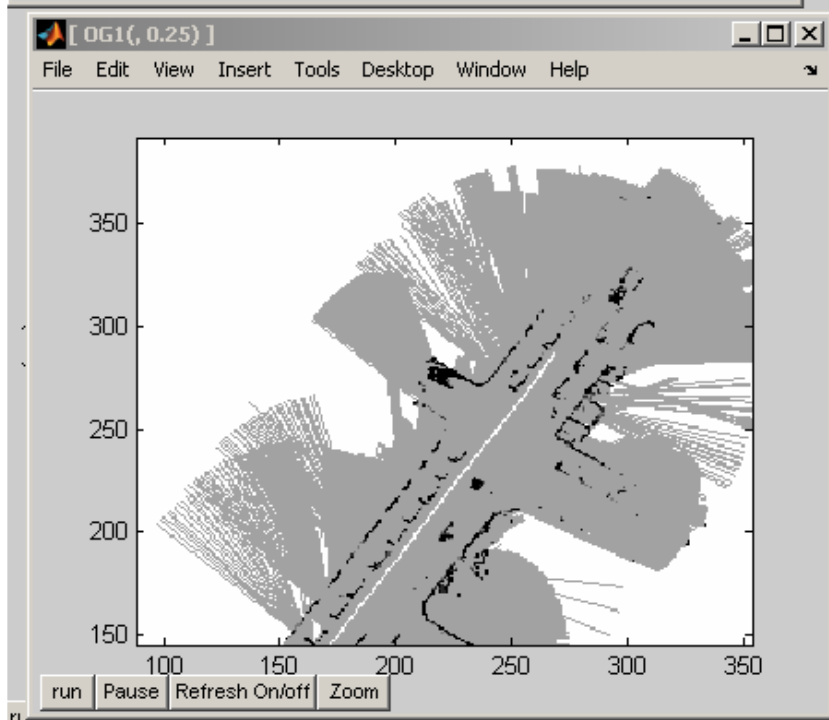


A 2D laser scan and its associated video image (from a video camera)
The sensors were installed at the front of a vehicle
(for a collision avoidance system)

By combining multiple scans, from different perspectives, mapping and 3D images can be obtained.

(these fusion processes require applying C.F. transformations!)





EXAMPLE IN OUR LECTURE: 2D SICK TIM

(WE SEE IT WORKING)



TIM561-2050101S80 | Part no.: 1106065
2D LiDAR sensors | TiM5xx

- **Application:** Outdoor
- **Integrated application:** Output of measurement data
- **Measurement principle:** HDDM
- **Working range:** 0.05 m ... 10 m
- **Angular resolution:** 0.33°
- **Digital outputs:** 1 (PNP, SYNC/device ready)
- **Connection type:** 1 x "Ethernet" connection, 4-pin M12 female connector, 1 x connection "Power/Synchronization output" 5-pin, M12 male connector, 1 x Micro USB female connector, type B

(HDDM: High Definition Distance Measurement = ToF approach)

Model 571 has range =25m
FoV=270 degrees.

<https://www.sick.com/au/en/detection-and-ranging-solutions/2d-lidar-sensors/tim5xx/c/g292754>

Let's see some LiDAR working , now.
(I have now a SICK TIM 541.)

And than, we end the lecture.